NFL Win/Loss Predictor

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CSCI 335 - RIT

Problem/Background

- Predicting NFL games is very difficult due to the high variance and constant changing of sports data
- A team's performance is never static and is heavily influenced by momentum, injuries, and other constantly changing attributes.
- Our objective is to build a robust model that accounts for temporal dependencies and non-linear interactions between team statistics

Why Care About the Problem

- Many shows centered around analysts trying to guess who will win a game
- If you are able to accurately predict games consistently then you could technically make money through betting
  - (disclaimer: this is not us encouraging you to go gambling)
- People enjoy debating one another on who will win a game and it is a reason some people enjoy sports.

Hypothesis

We believe that the Random Forest model and the Logistic Regression model will have the best performance since they seem to be the most common models that we saw use of in our research when looking at other peoples models.

Dataset Overview

- Data Sources: Aggregated 2024 NFL season data
  - Offense: Passing, Rushing, and Total Offense
  - Special Teams: Punting, Kicking, and Returns
  - Scoring & Results: Team Scoring Offense and the 2024 Season Schedule
- Data Volume: Processes all 272+ matches (Regular Season + Playoffs)
- Created a binary label ‘Home_Win’
- Structure: Wide dataset merging Home team stats and Away team stats onto every game row

Data Preprocessing & Feature Engineering

- Eliminating Data Leakage: Features use only information that is available before games start
- Time-Based Splitting: Instead of a random shuffle, following split used:
  - Train: Weeks 1–12
  - Validation: Weeks 13–14
  - Test: Week 15 through the Super Bowl
- Rolling Averages:
  - Calculated Season-to-Date
  - Last-5 Game means for critical metrics
- Matchup Difference: Created an advantage feature to show relative strength
- Schedule Fatigue: Rest Days between games for Short Weeks for Bye Weeks
- StandardScaler: Normalize features for models like Logistic Regression and MLP

Data Preprocessing (Cont)

Features (41 Total)

Pre-Kickoff Information Only:

- Context: Week, rest days, prior win %, games played
- Rolling Means (STD & Last-5): Points, yards, turnovers, wins per team
- Chronological Shift: One-game shift to eliminate data leakage
- Matchup Gaps: (Home - Away) calculated on last-5 stats

Training & Validation Strategy

Time-Based Split:

Phase 1: Training
Weeks 1 – 12

Phase 2: Validation (Optional)
Weeks 13 – 14

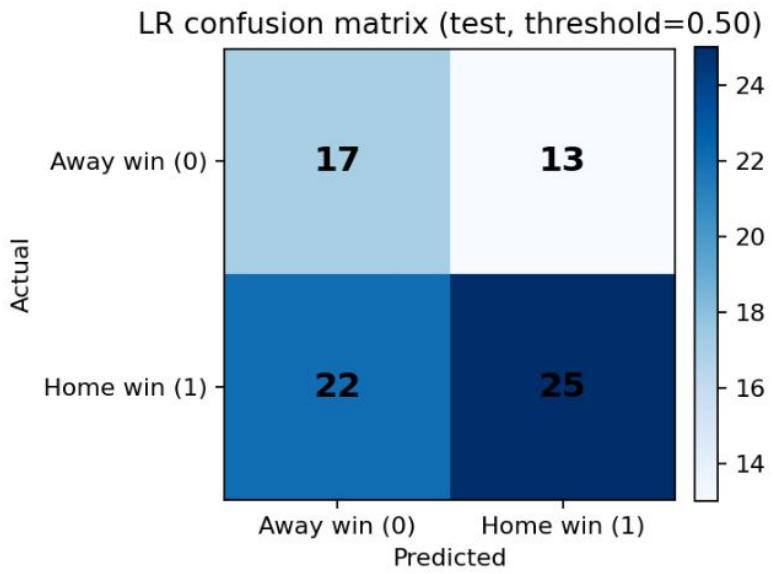
Phase 3: Testing
Week > 14 (Late Season + Playoffs)

Methods Used

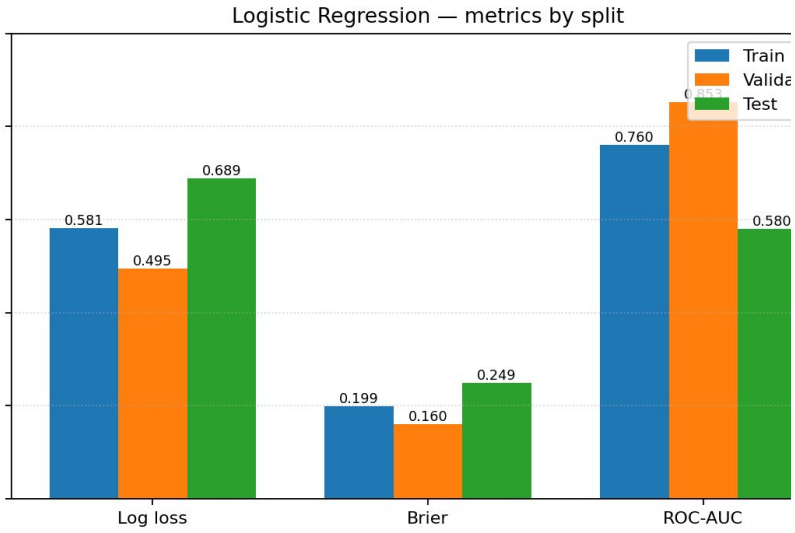
- Logistic Regression
- Random Forest
- KNN
- Multi-layer Perceptron (NN)

Logistic Regression Results

Confusion Matrix (Test Set)



Performance Metrics by Split



Logistic Regression Results (Cont)

Train Set

Accuracy

0.6927

Precision

0.6869

F1 Score

0.7120

Validation Set

Accuracy

0.8966

Precision

0.8889

F1 Score

0.9143

Test Set

Accuracy

0.5455

Precision

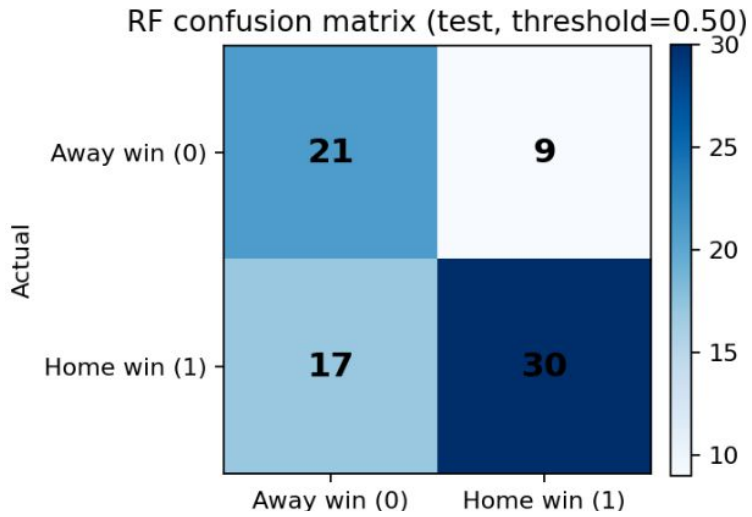
0.6579

F1 Score

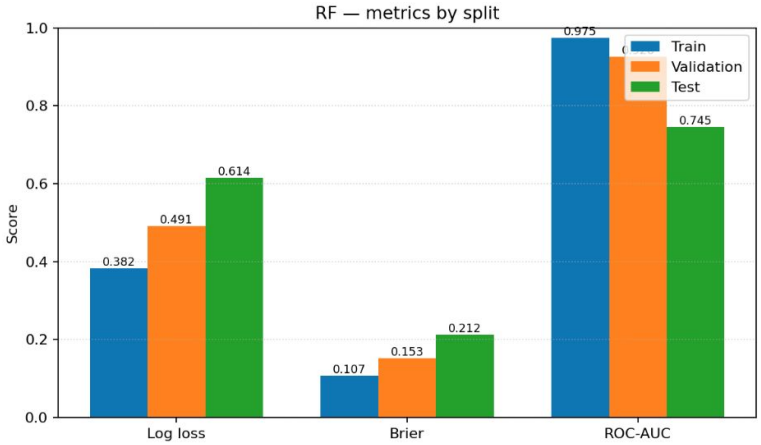
0.5882

Random Forest Results

Confusion Matrix (Test Set)



Performance Metrics by Split



Random Forest Results (Cont)

Train Set

Accuracy

0.9609

Precision

0.9293

F1 Score

0.9634

Validation Set

Accuracy

0.8621

Precision

0.8421

F1 Score

0.8889

Test Set

Accuracy

0.6623

Precision

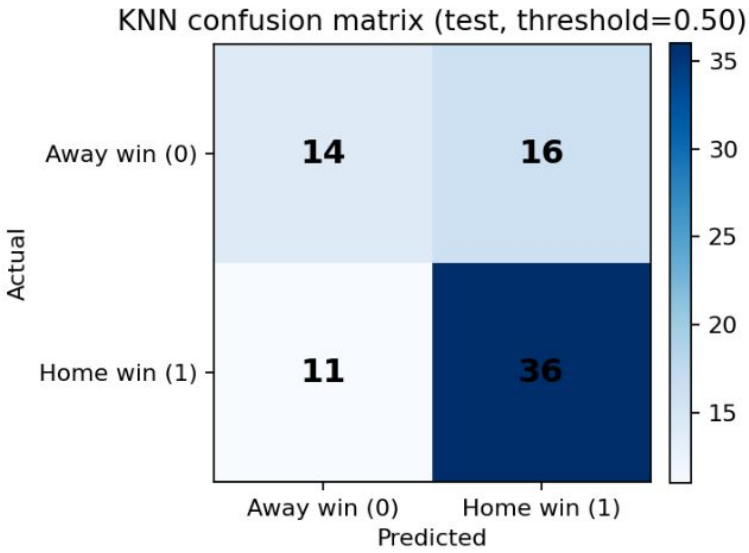
0.7692

F1 Score

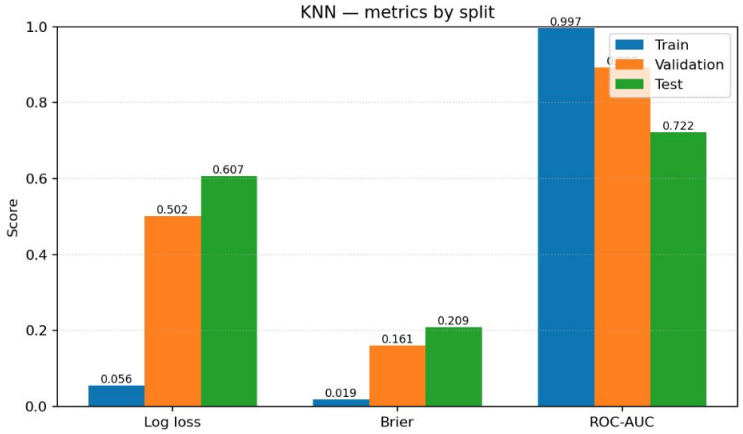
0.6977

KNN Results

Confusion Matrix



Performance Metrics by Split



KNN Results (Cont)

Train Set

Accuracy
0.9721

Precision
0.9485

F1 Score
0.9735

Validation Set

Accuracy
0.8276

Precision
0.8000

F1 Score
0.8649

Test Set

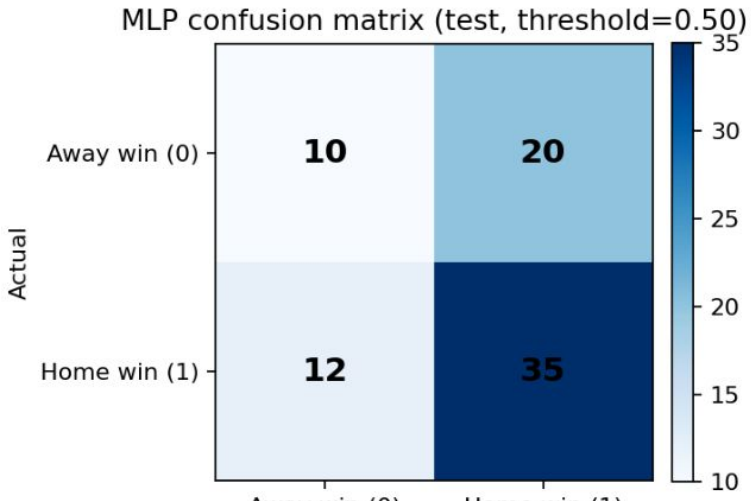
Accuracy
0.6494

Precision
0.6923

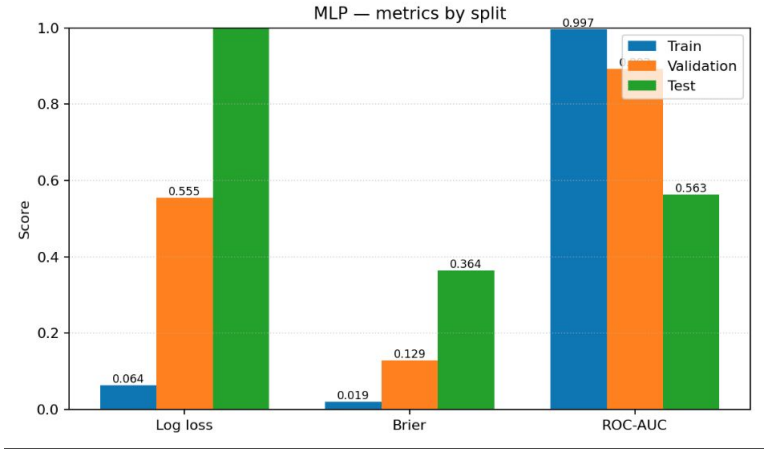
F1 Score
0.7273

Multi-Layer Perceptron (MLP) Results

Confusion Matrix



Performance Metrics by Split



MLP Results (Cont)

Train Set

Accuracy

0.9721

Precision

0.9485

F1 Score

0.9735

Validation Set

Accuracy

0.8621

Precision

0.8421

F1 Score

0.8889

Test Set

Accuracy

0.5844

Precision

0.6364

F1 Score

0.6863

Interpretations

- Our best 2 models were KNN and Random Forest
  - KNN most likely performed well since we had a relatively small data set
  - Random Forest most likely performed well since it is a form of ensemble learning that uses many trees to make a prediction.
- Worst model was the Logistic Regression
  - This could be because Logistic Regression assumes there is a linear relationship between the features and the outputs
- MLP had the most extreme overfitting
  - The model we made was too complex and needed hyperparameter tuning in order to not overfit as hard onto the training data

Limitations

- Amount of data used was relatively small as we only used 1 years worth of team stats
- Better hyperparamter tuning to facilitate better models
- Make use of some regularization methods to cut down on the overfitting that we saw occur
- Try splitting the data differently since all of the postseason games were in the test set and not every game has the same stakes which could affect the outcome of the game

Acknowledgements

https://pmc.ncbi.nlm.nih.gov/articles/PMC12463883/

https://www.sciencedirect.com/science/article/pii/S2772662224001413

Pro Football Stats, History, Scores, Standings, Playoffs, Schedule & Records | Pro-Football-Reference.com